1 Find Your Method

Use this decision tree to navigate to the chapter or case study most relevant to your research question.

flowchart TD A["What is yourresearch question?"] –> B["Map a research field"] A –> C["Evaluate researchperformance"] A –> D["Detect emergingtopics or trends"] A –> E["Study collaborationpatterns"] A –> F["Track open access& open science"]

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B --> B1["Bibliographic coupling<br>Ch. 16"]
B --> B2["Co-citation analysis<br>Ch. 16"]
B --> B3["Science mapping<br>Ch. 19"]
B --> B4["Topic modeling<br>Ch. 22"]
B --> B5["Case Study 1:<br>CRISPR Field Review"]

C --> C1["Author metrics<br>Ch. 10, 12"]
C --> C2["Journal metrics<br>Ch. 11"]
C --> C3["Institutional benchmarking<br>Ch. 13"]
C --> C4["Case Study 2:<br>Institutional Benchmark"]
C --> C5["Case Study 3:<br>Journal Portfolio"]

D --> D1["Keyword bursts<br>Ch. 25"]
D --> D2["Temporal analysis<br>Ch. 14"]
D --> D3["Embedding clustering<br>Ch. 24"]
D --> D4["Case Study 5:<br>Emerging Topics"]

E --> E1["Co-authorship networks<br>Ch. 15"]
E --> E2["Community detection<br>Ch. 18"]
E --> E3["Gender & equity<br>Ch. 28"]
E --> E4["Case Study 4:<br>Gender Gap"]

F --> F1["OA indicators<br>Ch. 27"]
F --> F2["Altmetrics<br>Ch. 26"]
F --> F3["Reproducibility<br>Ch. 33"]
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1.1 Quick reference table

Research question	Key chapters	Case study
Map a research field	16, 19, 22	CS 1: CRISPR
Evaluate an author	10, 12	—
Evaluate a journal	11	CS 3: Journal Portfolio
Benchmark an institution	13	CS 2: Institutional
Detect emerging topics	14, 24, 25	CS 5: Emerging Topics
Analyze collaboration	15, 18	—
Study gender gaps	28	CS 4: Gender Gap
Track open access	27	—
Build a dashboard	35, 37	—
Create a reproducible pipeline	34, 36	—